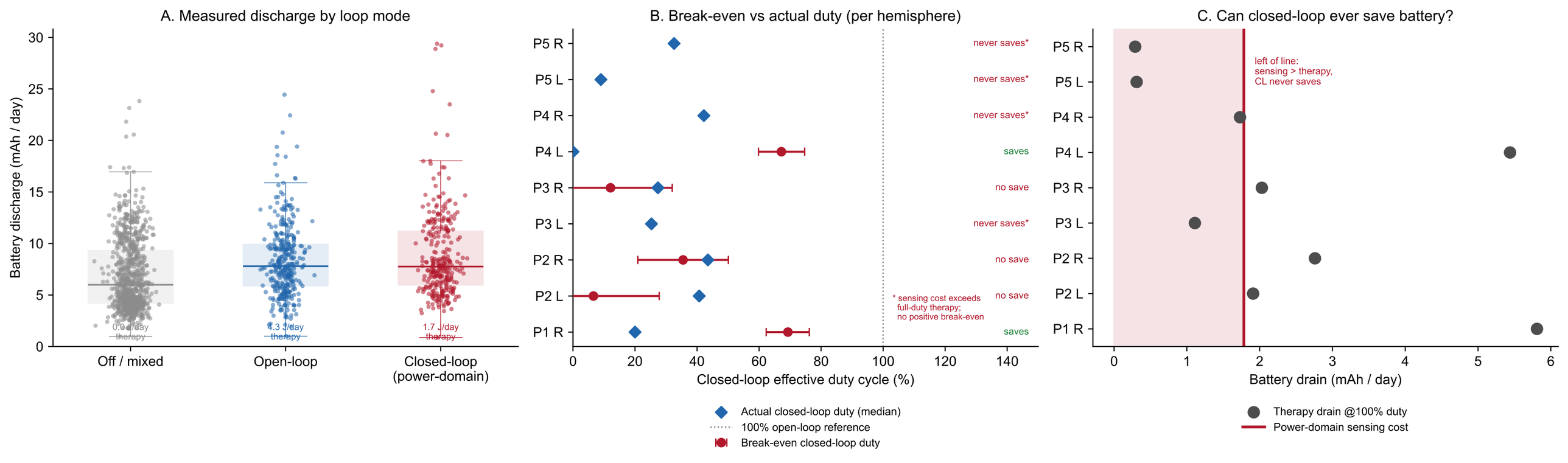
A. Measured discharge by loop mode
Battery discharge (mAh / day)
0
5
10
15
20
25
30
0.0 J/day therapy
4.3 J/day therapy
1.7 J/day therapy
Off / mixed
Open-loop
Closed-loop (power-domain)
B. Break-even vs actual duty (per hemisphere)
P5 R
P5 L
P4 R
P4 L
P3 R
P3 L
P2 R
P2 L
P1 R
never saves*
never saves*
never saves*
saves
no save
never saves*
no save
no save
saves
* sensing cost exceeds full-duty therapy; no positive break-even
0
20
40
60
80
100
120
140
Closed-loop effective duty cycle (%)
Actual closed-loop duty (median)
100% open-loop reference
Break-even closed-loop duty
C. Can closed-loop ever save battery?
P5 R
P5 L
P4 R
P4 L
P3 R
P3 L
P2 R
P2 L
P1 R
left of line: sensing > therapy, CL never saves
0
1
2
3
4
5
6
Battery drain (mAh / day)
Therapy drain @100% duty
Power-domain sensing cost